

Technology Management (MG2007) – Final Exam Answer Sheet

Q1(A) Tacit vs Explicit Knowledge

Tacit Knowledge: Personal, experience-based knowledge that is difficult to formalize and transfer.

Explicit Knowledge: Documented knowledge that can easily be stored and communicated.

Examples:

- Tacit: An engineer's troubleshooting skills.
- Explicit: A training manual or SOP document.

Conclusion: Tacit knowledge is harder to transfer but highly valuable, while explicit knowledge is easy to share and standardize.

Q1(B) Transactional vs Transformational Leaders

Transactional Leadership: Focuses on tasks, rewards, punishments, and maintaining existing systems.

Transformational Leadership: Focuses on inspiration, vision, innovation, and organizational change.

Examples:

- Transactional: Sales manager rewarding employees for meeting targets.
- Transformational: Elon Musk inspiring innovation at Tesla and SpaceX.

Q1(C) Product Road Mapping vs Technology Road Mapping

Product Roadmap: Focuses on future product features and releases.

Technology Roadmap: Focuses on future technologies and technical capabilities.

Examples:

- Product roadmap: Apple planning future iPhone features.
- Technology roadmap: Intel planning future processor technology.

Q1(D) ROI vs NPV

ROI (Return on Investment): Measures profitability percentage of an investment.

NPV (Net Present Value): Measures present value of future cash flows minus investment cost.

Conclusion: ROI is simple and fast, while NPV is more accurate because it considers time value of money.

Q2(A) Netflix's Freedom and Responsibility Philosophy

Netflix empowers employees with high freedom and responsibility. Employees can make independent decisions, encouraging creativity and experimentation.

Examples:

- A/B testing for recommendation systems.
- Continuous experimentation with user experience and content delivery.

This culture creates agility and supports continuous innovation.

Q2(B) How Netflix Maintains Innovation Leadership

Netflix maintains leadership through:

- Data analytics and AI.
- Cloud computing and scalable infrastructure.
- Netflix Originals and content innovation.
- Continuous experimentation and global expansion.

These strategies help Netflix remain competitive in the streaming industry.

Q2(C) Challenges of Such an Environment

Challenges include:

- Employee burnout due to high expectations.
- Job insecurity.
- Coordination problems among autonomous teams.
- Ethical and privacy concerns from heavy data collection.

Netflix must balance innovation with employee well-being and ethical practices.

Q3 Quantitative and Qualitative Project Selection Methods

Quantitative Methods:

- ROI
- NPV
- Payback Period
- Cost-Benefit Analysis

Qualitative Methods:

- SWOT Analysis
- Strategic Alignment
- Expert Judgment
- Risk Assessment

Businesses combine both methods to build strong project portfolios.

Q4 Ethical Issues in Innovative Industries

Major ethical issues include:

- Data privacy violations.
- AI bias and discrimination.
- Environmental impact and e-waste.
- Worker exploitation.
- Intellectual property misuse.
- Cybersecurity risks.

Ways to Be More Ethical:

- Transparent data policies.
- Ethical AI auditing.
- Sustainability initiatives.
- Fair wages and employee welfare.
- Compliance with laws and regulations.
- Ethics training programs.